

Event Information:

EventID :

139

Event Time :

Apr, 20, 2023, 09:42 AM

Rule :

SOC189 - VBScript Suspicious Behavior Detected

Level :

Incident Responder

Hostname :

David

IP Address :

172.16.17.31

Related Binary :

Purchase_Order.xls.vbs

Binary Path :

C:\Users\LetsDefend\Downloads\Purchase_Order\Purchase_Order.xls.vbs

Binary MD5 :

8FAF36EDFAE1EC0E8ECCD3C562C03903

Trigger Reason :

VBScript attempting to access sensitive system resources or files, such as the Windows Registry or system files, that are not related to its expected functionality.

Device Action :

Allowed

L1 Note :

When i searched the hash online the file seems it is some variant of wshrat. And it seems malicious. I'm assigning this alert for further investigations.

Analysis Steps:

Logs were searched with source IP 172[.]16.17.31 (David) in the relevant time-frame yielding 3 events. We can see what is likely C2 activity associated with the malicious file 103[.]47.144.80.

✓ 3 events (before Apr, 20, 2023, 09:47 AM UTC)

Event	
Field	Value
type	Firewall
source_address	172.16.17.31
source_port	49826
destination_address	103.47.144.80
destination_port	7045
time	Apr, 20, 2023, 09:47 AM

On VirusTotal the file is listed as malicious by 36/61 Vendors as of 11 days ago.

36
/ 61

Community Score

36/61 security vendors flagged this file as malicious

1c546d548bed639640efb02abd96013c33500172cfc0e8716c96bb196

Purchase_Order.als.vbs

vba malware direct-cpu-disk-access calls-wni runtime-modules long-sleeps persistence

Size

527.62 KB

Last Analysis Date

11 days ago

Reanalyze

Similar

More

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 34

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Code insights

The code is a VBScript designed to establish and maintain remote control over a compromised system. It employs an initial layer of obfuscation to hide its primary script content, which is deobfuscated during execution. Upon execution, the script performs the following core behaviors:
1. Persistence and Self-Copying: It copies itself to the temporary directory and the user's startup folder. It modifies HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run and HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run registry keys to ensure automatic execution on system startup.
[Show more](#)

Popular threat label

trojan.skid0x/heur2

Threat categories

trojan dropper worm

Family labels

skid0x heur2 skid0x

Security vendors' analysis

Do you want to automate checks?

AllCloud	Trojan.Win/Trojan.PMM	Antiy-AVL	Norm/Script.Agent
Arcabit	GT-VB.Heur2.Zbot.4.64C.D80BC	Avast	JS-Skiddo.A [Trj]
AVG	JS-Skiddo.A [Trj]	Avira (no cloud)	TR/JS.Skiddo.A
BitDefender	GT-VB.Heur2.Zbot.4.64C.D80BC	CTX	Vba.trojan.generic

In the behaviour section we can also identify the same C2 address seen in our logs.

Network Communication

HTTP Requests

+ POST http://chongmei33.publicvm.com:7045/is-ready

DNS Resolutions

+ chongmei33.publicvm.com

IP Traffic

TCP 46.246.86.2:7045 (chongmei33.publicvm.com)

TCP 103.47.144.80:7045 (chongmei33.publicvm.com)

Memory Pattern Urls

chongmei33.publicvm.com

On inspecting the endpoint sysmon logs we can see that the malicious file creates registry entries to maintain persistence via the use of the common LOLBin wscript.exe. The malicious file is scheduled to run every time the user logs in.

Operational Number of events: 6 (!) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	7/8/2026 9:16:07 AM	Sysmon	11	File created (r...
Information	7/8/2026 9:16:07 AM	Sysmon	13	Registry value ...
Information	7/8/2026 9:16:01 AM	Sysmon	11	File created (r...
Information	7/8/2026 9:16:01 AM	Sysmon	13	Registry value ...
Information	7/8/2026 9:15:56 AM	Sysmon	11	File created (r...
Information	7/8/2026 9:15:56 AM	Sysmon	13	Registry value ...

Event 13, Sysmon

General Details

EventType: SetValue
UtcTime: 2026-07-08 09:16:01.965
ProcessGuid: {a584806d-127c-6a4e-a100-000000005e01}
ProcessId: 6408
Image: C:\Windows\System32\wscript.exe
TargetObject: HKU\S-1-5-21-3163960855-2866672989-1813526453-1008\Software\Microsoft\Windows
\CurrentVersion\Run\Purchase_Order
Details: wscript.exe //B "C:\Users\LETSDE~1\AppData\Local\Temp\1\Purchase_Order.xls.vbs"
User: EC2AMAZ-ILGVOIN\LetsDefend

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon Logged: 7/8/2026 9:16:01 AM
Event ID: 13 Task Category: Registry value set (rule: RegistryEvent)
Level: Information Keywords:
User: SYSTEM Computer: EC2AMAZ-ILGVOIN
OpCode: Info
More Information: [Event Log Online Help](#)

It also replicates itself into the startup folder using the same tactic.

Event 11, Sysmon

General Details

RuleName: T1023
UtcTime: 2026-07-08 09:16:07.018
ProcessGuid: {a584806d-127c-6a4e-a100-000000005e01}
ProcessId: 6408
Image: C:\Windows\System32\wscript.exe
TargetFilename: C:\Users\LetsDefend\AppData\Roaming\Microsoft\Windows\Start Menu\Programs
\Startup\Purchase_Order.xls.vbs
CreationUtcTime: 2023-04-20 09:42:06.846
User: EC2AMAZ-ILGVOIN\LetsDefend

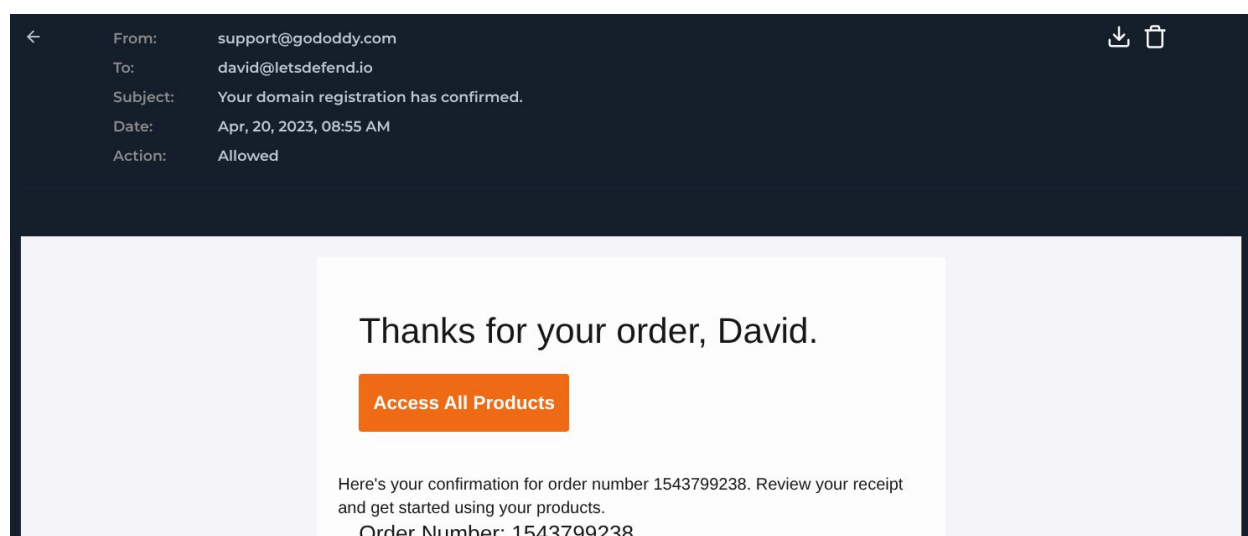
Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon Logged: 7/8/2026 9:16:07 AM
Event ID: 11 Task Category: File created (rule: FileCreate)
Level: Information Keywords:
User: SYSTEM Computer: EC2AMAZ-ILGVOIN
OpCode: Info
More Information: [Event Log Online Help](#)

The malicious file is seen to execute via heavily obfuscated visual basic code which masquerades itself as an Excel Spreadsheet.

```
Purchase_Order.xls - Notepad
File Edit Format View Help
vbCrLf & "v!.....!b?????C!.....!r?????L!.....!f?????P? & !.....!""?????P?!.!.....
vbCrLf & "v!.....!b?????C!.....!r?????L!.....!f?????P? & !.....!""?????P?!.!.....
vbCrLf & "v!.....!b?????C!.....!r?????L!.....!f?????P? & !.....!""?????P?!.!.....
!t?????P?????P?!.!.....!f?????P?????P?!.!.....!o?????P?????P?!.!1?????P
vbCrLf & "v!.....!b?????C!.....!r?????L!.....!f?????P? & !.....!""?????P?!.!e?
.....!""?????P?!.!""?????P?????P? & !.....!f?????P?????P?!.!.....!1?
P?!.!e?????P?!.!r?????P?!.!""?????P? & _ & _
vbCrLf & "v!.....!b?????C!.....!r?????L!.....!f?????P? & !.....!""?????P?!.!.....
vbCrLf & "v!.....!b?????C!.....!r?????L!.....!f?????P? & !.....!""?????P?!.!e?????P
vbCrLf & "v!.....!b?????C!.....!r?????L!.....!f?????P? & !.....!""?????P?!.!""?????P
vbCrLf & "v!.....!b?????C!.....!r?????L!.....!f?????P? & !.....!""?????P?!.!""?????P
vbCrLf & "d!.....!i?????P?!.!m?????P? !.....!b?????P?CR!.....!o?????P?!.!h?????P?!.
vbCrLf & "GIW!.....!o?????P?yZp!.....!f?????P?AYKDjWj!.....!c?????P?L!.....!o?????P?!.!
vbCrLf & "i!.....!i?????P?!.!n?????P?!.!m?????P?B!.....!s?????P?j!.....!r?????P?!.!
?????P?!.!""?????P?)" & _
vbCrLf & "OH!.....!f?????P?Y!.....!f?????P?M!.....!l?????P?SIqNA!.....!b?????P?!.!x????
vbCrLf & "GIW!.....!o?????P?yZp!.....!f?????P?AYKDjWj!.....!c?????P?L!.....!o?????P?!.!
vbCrLf & "OH!.....!f?????P?Y!.....!f?????P?MqNA!.....!b?????P?!.!x?????P?!.!c?????P
vbCrLf & "i!.....!i?????P?!.!n?????P?!.!m?????P?B!.....!s?????P?j!.....!r?????P?!.!
???P?), !.....!""?????P?!.!""?????P?)" & _
vbCrLf & "OH!.....!f?????P?Y!.....!f?????P?M!.....!l?????P?qNA!.....!b?????P?!.!x?????P
vbCrLf & "i!.....!e?????P?!.!x?????P?!.!e?????P?!.!c?????P?!.!u?????P?!.!
vbCrLf & "" & _
vbCrLf & ""
on error resume next
dim bCrOhcXWJZMKGZbWngJKnTGdWBxPUyWnOGhat
GIWoyZpfAYKDjWjcLomB = "?????P?GuoKQStZLxIUAgDxuMntqNAbxcLWfK1yKKUgrUTwUNLDIeXxbeNfPWlWqed"
inmBsJrrYiGcfxeBXSCJ = replace(inmBsJrrYiGcfxeBXSCJ,split(GIWoyZpfAYKDjWjcLomB,"GuoKQs")(0), "")
OHfYfM1SIqNAbxcLWfK1yKKUgrUTwUNLDIeXxbeNfPWlWqedYrKWLCLWkwR
GIWoyZpfAYKDjWjcLomB = "!.....!zLdQWiaUfOuzDSjKYcqNAbxcLWfK1yKKUgrUTwUNLDIeXxbeNfPWlWqedXp"
OHfYfMqNAbxcLWfK1yKKUgrUTwUNLDIeXxbeNfPWlWqed1SIYrKWLCWkwR
inmBsJrrYiGcfxeBXSCJ = replace(inmBsJrrYiGcfxeBXSCJ,split(GIWoyZpfAYKDjWjcLomB,"zLdQWiaUfO")(0), "")
OHfYfM1qNAbxcLWfK1yKKUgrUTwUNLDIeXxbeNfPWlWqedSIYrKWLCWkwR
execute inmBsJrrYiGcfxeBXSCJ
```

The last line is `execute inmBsJrrYiGcfxeBXSCJ`. This tells the system to take all the scrambled text, decode it in memory, and execute it as a command.

It can be observed that the malicious file was originally delivered via spearphishing email posing as a godaddy domain purchase receipt coming from the spoofed address support@[godaddy].com



The C2 IP is not in our CTI database and is external to the organisation. No emails have been received indicating penetration testing involving the IP for the relevant time period. VirusTotal indicates that it is a Pakistani address associated with 103[.]47.144.80 AS 205009 (HDM Solutions LTD). 1/91 security vendors flagged this IP address as malicious.

1

Community Score

1/91 security vendor flagged this IP address as malicious

103.47.144.80 (103.47.144.8/24)
AS 205009 (HDM Solutions LTD)

PK

Last Analysis Date
1 month ago

ReanalyzeMore

DETECTIONDETAILSRELATIONSCOMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis

Do you want to automate checks?

alphaMountain.ai	Malicious	Abusix	Clean
Acronis	Clean	ADMINUSLabs	Clean
AliLabs (MONITORAPP)	Clean	AlienVault	Clean
Antiy-AVL	Clean	BitDefender	Clean

On AbuseIPDB this IP was reported 2 times with a 0% confidence of abuse. They state that it is a Dutch address.

103.47.144.80 was found in our database!

This IP was reported 2 times. Confidence of Abuse is 0%?

0%

ISP

Cyber-Security-SG

Usage Type

Data Center/Web Hosting/Transit

ASN

[AS205009](#)

Domain Name

pointtoserver.com

Country

 Netherlands

City

Amsterdam, North Holland

Based on following Playbook procedure and the analysis of available information, this alert is determined to be a successful True Positive.

Report:

Incident Summary

Analysis confirms a True Positive WSHRAT (Windows Script Host RAT) infection on the host 'David'. The initial access vector was a spearphishing email masquerading as a GoDaddy domain receipt, delivering an obfuscated VBScript disguised with a double extension (.xls.vbs). The payload successfully executed, established persistence using native Windows binaries (LOLBins), and initiated C2 communications.

Incident Details

- **Alert Name:** SOC189 - VBScript Suspicious Behavior Detected
- **Event Time:** Apr, 20, 2023, 09:42 AM
- **Target Host:** David (172[.]16.17.31)
- **File Name:** Purchase_Order.xls.vbs
- **File Hash (MD5):** 8FAF36EDFAE1EC0E8ECCD3C562C03903
- **C2 Address:** 103[.]47.144.80:7045

Analysis and Evidence

- **Delivery:** A phishing email from support[@]gododdy[.]com containing the malicious attachment successfully bypassed filters and was permitted by the endpoint action Allowed.
- **Execution:** The file is a heavily obfuscated VBScript. Upon execution, it decodes the payload in memory (execute inmBsirrYiGcfxeBXSCJ).
- **Persistence:** Sysmon Event ID 13 confirms the malware utilised wscript.exe to modify the Windows Registry, placing a persistence mechanism in the HKU\...\CurrentVersion\Run key. Sysmon Event ID 11 confirms it also copied itself to the user's Startup folder to guarantee execution upon reboot.
- **OSINT:** VirusTotal flags the hash (36/61) as a WSHRAT/Zbot variant. Firewall logs confirm the host established outbound connections to 103[.]47.144.80 over port 7045. OSINT confirms this IP matches the C2 infrastructure associated with this payload.

Actions Taken

1. Removed malicious email from inbox.
2. Terminated malicious wscript.exe processes, remove the Registry Run key, deleted the payload from the Temp and Startup directories, and globally blocked the C2 IP.
3. Closed the alert as a True Positive.

Let's Defend Result

SOC189 - VBScript Suspicious Behavior Detected

True Positive (+5 Point)

Does the device need to be isolated? (+5 Point)

Which technique was used in the credential access tactic? (+5 Point)

Were privilege escalation techniques used in the attack? (+5 Point)

What is the persistence method used in the attack? (-5 Point)

What is the initial access method used in the attack? (+5 Point)

Has the malware been executed on the device? (+5 Point)

What type of malware is used in the attack? (+5 Point)

I chose scheduled task as the persistence method but since the malware uses both a scheduled task and copies itself to the startup folder it must have expected the startup file as the correct answer.